SECTION 1 · SUBSECTION 1.2
SYSTEM PURPOSE AND BOUNDARIES

1.2.1 INTRODUCTION.

Project Gnosis converts an unfamiliar public GitHub repository into a reading aid. A user submits a repository address; the system fetches a filtered copy of its text files, parses them into symbols, measures their structural weight, and returns a document that names the entry points, the risky files and a safe reading order. The system never executes the target repository (figure 1.2-1).

The boundary is deliberately narrow. Gnosis reads source text and repository metadata. It does not build, run, test, patch or deploy the analysed project, and it cannot see a private repository unless the caller supplies a token through the API. Its output is a snapshot of one branch at one commit; it does not update when the repository changes.

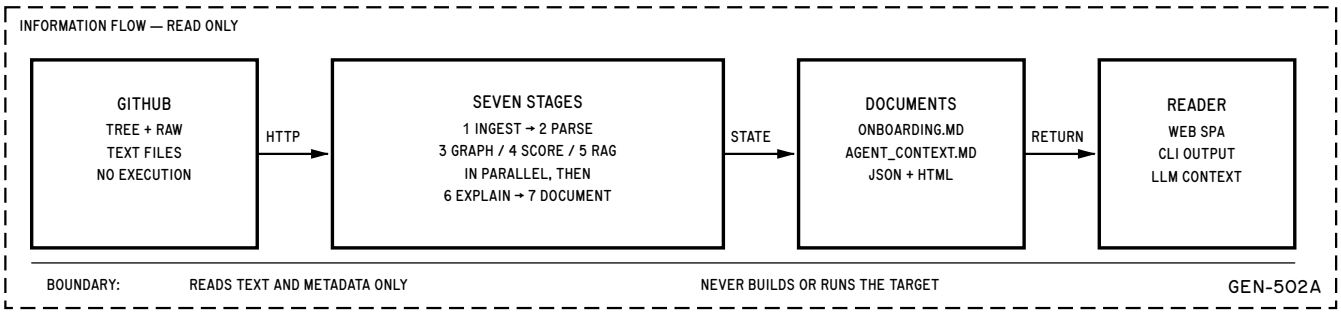


Figure 1.2-1. Input, Seven-Stage Transformation and Output

1.2.2 FUNCTIONAL DESCRIPTION.

Stage 1 lists every path on the default branch, keeps recognised source and configuration extensions, then fetches the raw text of each file. Stage 2 parses each file with a tree-sitter grammar and records functions, classes, imports and docstrings. Stages 3, 4 and 5 run concurrently: the graph resolves internal imports, the scorer computes cyclomatic complexity, and the retrieval stage chunks and embeds the code. Stage 6 selects the most important files and asks a hosted model to explain them; stage 7 assembles the results into Markdown and JSON (figure 1.2-2).

1.2.2.1 Repository Sizing.

A repository is classified by the number of files that survive filtering. At most 300 files is FULL; up to 1,000 is FULL (WARNING); up to 3,000 is SAMPLED. The API rejects anything above 3,000 before a job is created; the CLI caps the manifest at 3,000 and continues. Sorting is by path depth, then alphabetically, so shallow files survive a cap first.

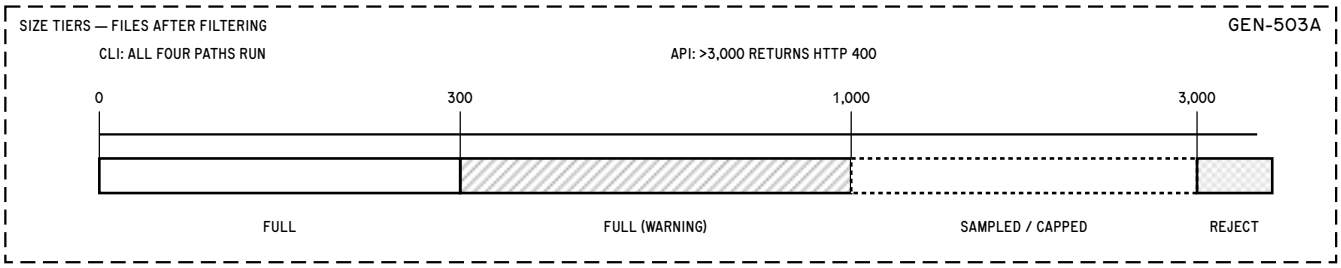


Figure 1.2-2. Repository Size Tiers and Enforcement Point (band widths not to scale)

1.2.3 PERFORMANCE AND DESIGN DATA.

Table 1.2-1. Boundary and Sizing Parameters

Parameter	Value
Full-analysis ceiling	300 files (env MAX_FULL_ANALYSIS_FILES)
Warning ceiling	1,000 files (MAX_WARNING_ANALYSIS_FILES)
Sampled / ingestion cap	3,000 files (MAX_SAMPLED_ANALYSIS_FILES)
Per-file size ceiling	150,000 bytes
Accepted extensions	18, from .py to .md
Explanation cap	20 files default; 0–100 allowed
Output languages	8 parsed; 8 scored
Repository write access	None

CAUTION

A sampled run is not a whole-repository verdict. Only the first 3,000 filtered paths, sorted by depth, are read. Files that are not ingested cannot appear in the graph, the scores or the document. The header records the mode; the reader must not generalise from a sample.

SCOPE CHECK	REVIEWER
1 CONFIRM - public URL, one branch	
2 CONFIRM - text only, no target build	
3 CONFIRM - no private repo without token	
√ MODE - Full, Warning or Sampled	
4 CONFIRM - output is a snapshot	